

Writing Systems in East and Southeast Asia

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based on Goddard Chapter 6

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Introduction to Writing Systems

- Language: Primarily spoken, but writing is a stable, visible form.
- Writing is integral to language culture.
- Study of writing systems reveals cultural history and linguistic insights.
- Two basic principles: logographic and phonographic.
- Logographic: Symbols represent words/morphemes.
- Phonographic: Symbols represent sounds (phonemes/syllables).

Types of Writing Systems

- Logographic Systems:
 - Symbols represent morphemes (e.g., Chinese, Hieroglyphics).
- Phonographic Systems:
 - Alphabetic (segmental): Symbols represent phonemes (e.g., Latin script).
 - Syllabic: Symbols represent syllables (e.g., Japanese kana).
 - Featural: Symbols represent phonetic features (e.g., Korean Hangul).
- Mixed Systems:
 - Combination of logographic and phonographic elements (e.g., Japanese).

Alphabetic Systems: Malay/Indonesian — Jawi

- Traditional Jawi script:

- Arabic origin.
- Right to left writing.

- Example:

(1) السلام عليكم *as-salām 'alaykum* “Peace be upon you.”

- Jawi (جاوي) is a shortening of الجزائر الجاوي, *Al-Jaza'ir Al-Jawi* “Java Archipelago”
- *Jawi* is often used to mean ‘Malay’, for example in *Bahasa Jawi* “Malay Language”
- Jawi has 31 original Arabic letters;
 - 6 new letters constructed to fit phonemes native to Malay: ca (چ < /tʃ/), nga (ڠ < /ŋ/), pa (ڤ < /p/), ga (ڬ < /g/), va (ڤ < /v/),
 - 1 additional phoneme used in foreign loanwords: nya (ڬ < /n/).

Jawi Script Details

- Jawi script is highly stylized for brush writing.
- Letters change form based on their position in the word.
- Example of a Jawi phrase:
(2) بالي جكرتا (Bali Jakarta) - Bali to Jakarta
- Usage has declined but still part of Malaysian education.
- Used in religious documents and decoration

وورما نيت بهاس

Wordnet Bahasa

Alphabetic Systems: Malay/Indonesian — Rumi

- Introduced by colonial powers.
 - Malay spelling followed English
 - Indonesian spelling followed Dutch
- Standardized in 1972 for both Malaysia and Indonesia.
 - *di-buat* → *dibuat*, *di-rumah* → *di rumah*
 - *ch* and *tj* /tʃ/ → *c*: *ketjap/kechap* → *kecap*
 - *njonja/nyonya* → *nyonya*
 - *ke-barat²-an* → *kebarat-baratan* “westernized”

Alphabetic Systems: Thai

- Derived from Old Khmer script.
Derived from Mon, originally Indic
- No spaces between words; consonants represented by main letters, vowels by diacritics.
- Example:
(3) สวัสดี (sawasdee) - Hello
- Complex relationship between letters and tones.

Thai Tones and Consonant Classes

- Consonant letters divided into three classes: high, middle, and low.
- Tone determined by initial consonant, tone marks, and vowel length.
- Thai words with initial /kh/ (Smalley 1994: 184)

Thai Word	Meaning	Thai Word	Meaning
ขา	khǎa leg	คา	khaa embedded
ข่า	khàa galangal	ค่า	khâa cost
ข้า	khâa slave	ค้า	kháa trade
ขาด	khàat lack	คาด	khâat to strap
ขัด	khàt obstruct	คัด	khát be clogged

- Complicated function of the initial consonant, diacritic and final consonant
- But generally deterministic
- Like most writing systems, sound changes quicker than writing

Thai vowels

อะ

a

[a]

อ

a

[a]

อา

a

[a:]

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Thai diphthongs

เียะ	เีย	เือะ	เือ	ัวะ	ัว	ว
ia	ia	uea	uea	ua	ua	ua
[iaʔ]	[ia]	[ueaʔ]	[uea]	[uaʔ]	[ua]	[ua]
ิว	เียว	เวย	เวย	เวย	อว	เียว
io	eo	eo	aeo	ao	ao	iao
[iu/iw]	[eu/ew]	[e:u/e:w]	[ɛ:u/ɛ:w]	[au/aw]	[a:u]	[iau/iaw]
้าย	ไอ	ไอ	ไวย	อาย	อ้อย	ออย
ai	ai	ai	ai	ai	oi	oi
[ai/aj]	[ai/aj]	[ai/aj]	[ai/aj]	[a:i/a:j]	[ɔi/ɔj]	[ɔ:i/ɔ:j]
ไอย	อุย	เอย	อวย	เือย	อำ	ฤ
oi	ui	oei	uai	ueai	am	rue
[o:i/o:j]	[ui/u:j]	[ɤ:i/ɤ:j]	[uai/uaj]	[uai/uaj]	[am]	[ruu/ri]
ฤา	ร	ร	อ	อ		
rue	lue	lue	ʔ	silences		
[ruu:]	[luu]	[luu:]	[~]	consonant		

Korean Hangul

한글

han

geul

ㄱ ㄴ ㄷ ㄹ ㅁ ㅂ ㅅ ㅇ ㅈ ㅊ

g n d r m b s -ng j ch

크 트 표 흥 ㅏ ㅑ ㅓ ㅕ ㅗ

k t p h a eo o u eu i

Image from Wikipedia (2024)

Featural Systems: Korean Hangul

- Invented by King Sejong (15th century)
 - the process of invention is well documented in the Hunmin jeongeum, or “The correct sounds for the instruction of the people”
- Represents phonemes, arranged in syllable blocks.
- Example:

(4) 안녕하세요 *annyeonghaseyo* “hello”
- It was originally looked down on by scholars (who use hanzi) as used by women, children and the uneducated
- It was not widely adopted until the 19th century

Korean Hangul: Design Principles

- Vowels represented by distinct vertical/horizontal strokes.
- Consonant shapes reflect place of articulation.
- One syllable is one block

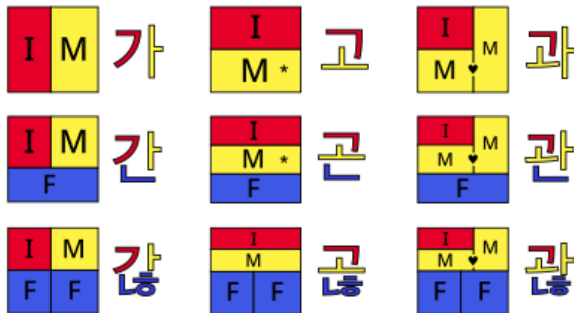
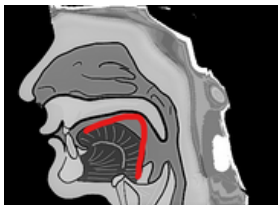


Image from Wikipedia (2024), cropped.

Korean Consonants: shape $\approx$ articulation



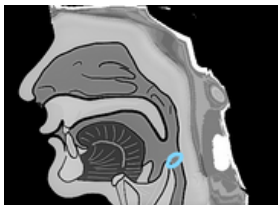
ㄱ /k/



ㄴ /n/



ㅅ /s/



ㅇ /ng/



ㅁ /m/

Images from Wikipedia (2024)

Korean Consonants: shape $\approx$ articulation

	Basic	Simple	Aspirated	Tense
Velar		ㄱ (k)	ㅋ (k ^h)	꺄 (k̚)
Fricatives		ㅅ (s)		ㅆ (s̚)
Palatal		ㅈ (tɕ)	ㅊ (tɕ ^h)	ㅉ (tɕ̚)
Coronal	ㄴ (n)	ㄷ (t)	ㅌ (t ^h)	ㄸ (t̚)
Bilabial	ㅁ (m)	ㅂ (p)	ㅍ (p ^h)	ㅃ (p̚)
Dorsal	ㅇ (ʔ)	ㅇ (ŋ)	ㅎ (h)	

Summary

- Hangul is a well-motivated alphabetic system, not a direct mapping of phonetic features
- Many sounds change in combination, so technically it is morphophonemic
- It is said that “A wise man can acquaint himself with them before the morning is over; even a stupid man can learn them in the space of ten days.”
Hunmin Jeongeum Haerye, postface of Jeong Inji, p. 27a, translation from Gari K. Ledyard, *The Korean Language Reform of 1446*, p. 258

Cia-Cia

- Cia-Cia (Bahasa Ciacia / 바하사찌아찌아) is a member of the Celebic branch of the Malayo-Polynesian language family (Austronesian)
- Spoken by about 80,000 people on Buton Island, a part of Indonesia to the south east of Sulawesi
- It used to be written in a Jawi-like script called Gundhul and is now mainly written in Rumi
- In 2007 Professor Chun Tai-hyun, a professor of Malay and Indonesian linguistics at Hankuk University of Foreign Studies in Seoul developed a Hangeul orthography
 - For fricative /v/ it uses the obsolete jamo **ㅞ** *kapyeounpieup* ‘light pieup’
- It was privately taught since 2009 using a textbook written by Lee Ho-young, a linguistics professor at Seoul National University.
- It was abandoned around 2012, and revived around 2022
- A Cia-Cia dictionary in Hangul was published in December 2021.
- Hangul remains in use in schools and on local signs

Cia-cia being used



From Language of the Month November 2024 –Cia-Cia By L S, The National Museum of Language (online virtual museum)

Chinese Writing System

- Ancient origins, evolved from pictograms.
- Continually in use for over 3,000 years
- A powerful force for cultural and political unification
- Also adopted by surrounding languages (Japanese, Korean, Vietnamese)
- Characters mainly represent morphemes, often compound structures.
- Example:
 - (5) 你好 *nǐ hǎo* “hello ‘you good’ ”
 - (6) 山 *shān* “mountain”
- Reformed to simplified characters in the 1950s

Development of Chinese character horse 马

Type of Characters	Descriptions
金文 文	Bronze script Jin wen 15th - 11th centuries B.C.E.
甲骨文 文	Oracle-bone script Jia gu wen 12th - 11th centuries B.C.E.
大篆 篆	Large-seal script da chuan c. 8th century B.C.E.
小篆 篆	Small-seal script xiao chuan 2nd century B.C.E.
隸書 書	Clerical script li shu 2nd century C.E.
楷書 書	Standard script k'ai shu since c. 4th century C.E.
行書 書	Running script Xing shu since c. 4th century C.E.
草書 書	Cursive script Chao shu since c. 4th century C.E.

Logograph writing system example: Chinese

Pictographs

女 子 口 日 月 山 川 豕 目 心 雨 田 木 龜
woman child mouth sun moon mountain river pig eye heart rain field tree turtle

Ideographs

一	二	三	上	下	中	力	凸	凹
one	two	three	above	below	middle	strength (plough)	convex	concave

Compound Pictographs / Compound Ideographs

好	安	明	家	思	牢	雷	男
good	peaceful	bright	home/family	thought	prison	thunder	man/male
(woman + child)	(woman under a roof)	(sun + moon)	(pig under a roof)	(heart + field)	(cow under a roof)	(rain cloud over a field)	(field + strength)

Semantic-phonetic compounds

Semantic-phonetic compounds

	phonetic component				
	古 gǔ	扁 biǎn	敖 áo	旁 páng	堯 yáo
semantic component (radical)	人 (person)	偏 piān	傲 ào	傍 bāng	僥 jiào
	言 (words)	諛 xiǎn	謗 bàng	謗 bàng	謗 bàng
	虫 (insect)	蝙 biān	螫 shí	螃 páng	蟯 xiào
	金 (metal)	鈷 gǔ	鑿 áo	鎊 bàng	鎊 bàng
	估 gū	編 biān	鰲 áo	磅 bàng	鎊 bàng
	(to guess)	(biased)	(proud)	(beside)	(lucky)
	話 huà	辯 biàn	謗 bàng	謗 bàng	謗 bàng
	(commentaries)	(to quibble)	(to slander)	(to libel)	(to argue)
	蛄 gū	蝙 biān	螫 shí	螃 páng	蟯 xiào
	(mole cricket)	(bat)	([crab/s] nippers)	(crab)	(worm)
	鈷 gǔ		鰲 áo	鎊 bàng	鎊 bàng
	(cobalt)		(griddle)	(pound sterling)	(cymbals)

97% of Chinese characters are phonetic compounds (Sproat 2010)! Other estimates are lower (Wang, pc 2013)

Structure of Chinese Characters

- Phonetic component gives clues to pronunciation.
- Semantic radical provides meaning hints.
- Characters require memorization due to irregular phonetic-semantic correspondences.

Using Chinese Characters for Non-Mandarin Chinese

- Chinese characters were designed for Mandarin, which poses challenges for other Sinitic languages.
- Problem: Phonetic clues in Mandarin characters often don't apply to languages like Cantonese and Taiwanese.
- For literacy in Cantonese or Taiwanese, familiarity with Mandarin's written form is often assumed.
- Despite differences, some words in Cantonese and Taiwanese can be represented with Mandarin characters.
- Issue: Many words and grammatical elements in other Sinitic languages lack standard Mandarin equivalents.

Writing Strategies for Cantonese

- Cantonese writers use various strategies to adapt characters:
 - ➊ **Unique Cantonese Characters:** Characters created for words without Mandarin equivalents.
 - ➋ **Phonetic Resemblance:** Using Mandarin characters that sound similar to Cantonese words.
 - ➌ **English Letters:** Borrowing letters to represent Cantonese sounds.
- Example of phonetic adaptation: The Cantonese plural suffix **-deih** is represented by combining ‘mouth’ and ‘earth’ characters, resembling the pronunciation **deih**: 㗎

Workarounds

(7) 你 喺 嗰度 好 喇, 千 祈 咪 搞 佢 啲
 你 o 係 果度 好 la, 千 祈 咪 搞 佢 D

you being there good SFP, thousand pray don't mess with he/she of

嘢 o

野 o

things/stuff .

You'd better stay there, and under no circumstances mess with his/her stuff.

- use 'o' instead of 口
- use alphabet: 'D'
- use homophones: 果度 for 嗰度
- because Cantonese is mainly spoken, there is considerable variation in the orthography

Loan words

Because of the influence of English, Cantonese tends to use more loan words (as does Singapore Mandarin). Note that speakers also freely use Mandarin vocabulary, ...

Cantonese	Jyutping	English	Mandarin	Pinyin
BB	bi4 bi1	baby	嬰兒	ying1 er2
巴士	baa1 si2	bus	公交车	gong1 jiao1 che1
拜拜	baai1 baai3	bye bye	再见	zai4 jian4
沙展	saa1 zin2	sergeant	排长	pai2 zhang3

Japanese Writing System

- Combines kanji (logographic) and kana (syllabic).
- Example:

(8) こんにちは konnichiwa (hello)
 山 yama (mountain)

Syllabaries: Hiragana and Katakana

- Hiragana: Used for particles, auxiliary verbs, and suffixes.
 - Derived from cursive versions of kanji with suitable pronunciation
- Katakana: Used for loanwords, mimetics, and emphasis.
 - Derived from components of kanji with suitable pronunciation

Hiragana - a true syllabic alphabet

平仮名 (ひらがな) hiragana

a	あ	安	i	い	以	u	う	宇	e	え	衣	o	お	於
ka	か	加	ki	き	幾	ku	く	久	ke	け	計	ko	こ	己
sa	さ	左	shi	し	之	su	す	寸	se	せ	世	so	そ	曾
ta	た	太	chi	ち	知	tsu	つ	川	te	て	天	to	と	止
na	な	奈	ni	に	仁	nu	ぬ	奴	ne	ね	祢	no	の	乃
ha	は	波	hi	ひ	比	fu	ふ	不	he	へ	部	ho	ほ	保
ma	ま	末	mi	み	美	mu	む	武	me	め	女	mo	も	毛
ya	や	也				yu	ゆ	由				yo	よ	与
ra	ら	良	ri	り	利	ru	る	留	re	れ	礼	ro	ろ	呂
wa	わ	和	wi	ゐ	為				we	ゑ	恵	wo	を	遠
												n	ん	无

Katakana - a true syllabic alphabet

片仮名 (カタカナ) katakana

a	ア	阿	i	イ	伊	u	ウ	宇	e	エ	江	o	オ	於
ka	カ	加	ki	キ	幾	ku	ク	久	ke	ケ	介	ko	コ	己
sa	サ	散	shi	シ	之	su	ス	須	se	セ	世	so	ソ	曾
ta	タ	多	chi	チ	千	tsu	ツ	川	te	テ	天	to	ト	止
na	ナ	奈	ni	ニ	二	nu	ヌ	奴	ne	ネ	祢	no	ノ	乃
ha	ハ	八	hi	ヒ	比	fu	フ	不	he	ヘ	部	ho	ホ	保
ma	マ	万	mi	ミ	ミ	mu	ム	牟	me	メ	女	mo	モ	毛
ya	ヤ	也				yu	ユ	由				yo	ヨ	與
ra	ラ	良	ri	リ	利	ru	ル	流	re	レ	礼	ro	ロ	呂
wa	ワ	和	wi	ヰ	井				we	エ	恵	wo	ヲ	乎
												n	ン	无

Extending Hiragana and Katakana

- **Dakuon** (濁音: ˊ): voicing
 - か (ka) → が (ga);
 - さ (sa) → ざ (za);
 - た (ta) → だ (da);
 - ハ (ha) → バ (ba)
- **Handakuten** (半濁点: °) plosive
 - は (ha) → ぱ (pa)
- **Yōon** (拗音): palatalized
 - Consonant + 'i' combines with small "ya," "yu," or "yo" (や, ゆ, よ).
 - き (ki) + や (xya) → きゃ (kya);
 - し (shi) + よ (xyo) → しょ (sho); ...
- **GōYōon** (合拗音): labialized (now only in Okinawan)
 - くわ (kwa), くゐ (kwi), くゑ (kwe), くを (kwo)
- **Sokuon** (促音):
 - Small つ marks a geminate consonant normally transliterated by doubling さっか *sakka*

Kanji and Mixed System Writing

- Lexical roots written in kanji.
- Grammatical elements in hiragana.
- Foreign words in katakana (or romaji).
- Mixed system allows nuanced text creation.

(9) 彼女は Tシャツを 着.ている
 kanojo ha T sha-tu wo ki te iru
 she TOP T shirt ACC wear PRG
 She is wearing a T-shirt

Homophones and Context Dependence

- Japanese has many homophones due to the limited set of sounds in the language.
- Kanji helps distinguish words with the same pronunciation but different meanings, yet:
 - Words with identical readings may use different kanji.
 - Context often determines which kanji is appropriate, adding complexity.
- Example: こうしょう (kōshō) can mean "negotiation" (交渉) or "factory" (鋳床), depending on kanji and context. (WWWDIC has 44 possible entries)

Why Kanji Have Multiple Readings

- Kanji have multiple readings due to their integration from Chinese into Japanese over centuries.
- Japanese and Chinese are linguistically different, so adaptations were necessary.
- Kanji readings can be divided into two main types:
 - **On' yomi** (音読み): Chinese-derived readings.
 - **Kun' yomi** (訓読み): Native Japanese readings.
- This dual system has led to multiple ways of reading kanji, depending on context.

Historical Borrowing from Chinese

- Kanji were imported from China starting in the 5th century, with Chinese pronunciations.
- **On' yomi** (音読み) readings were derived from Chinese pronunciations.
- Different periods and regions of Chinese influence led to varied on' yomi.
- For example, 山 has the reading **さん** ("san") in compounds, like in 火山 ("kazan," meaning "volcano").
- The Chinese-derived readings are often used in compound words.

Native Japanese Words (Kun' yomi)

- Many kanji represented concepts that already had native Japanese words.
- To adapt, kanji were assigned **kun' yomi** (訓読み), or native Japanese readings.
- Example: 明 can be read as あか in the native Japanese word 明るい *akarui* “bright”.
- Kun' yomi is often used when kanji stand alone or in names.
- This creates another layer of readings that are essential for native vocabulary.
- Many characters also have readings normally only used in names **nanori** (名乗り)

Multiple Chinese Influences: Go-on, Kan-on, Tō-on

- Kanji were imported from China starting in the 5th century, with Chinese pronunciations.
- **On' yomi** (音読み) readings were derived from Chinese pronunciations.
- Kanji were borrowed in waves, reflecting different Chinese dynasties and regions:
 - **Go-on** (呉音): Earlier readings from the Wu region.
 - **Kan-on** (漢音): Pronunciations from the Tang Dynasty.
 - **Tō-on** (唐音): Later readings from the Song and Ming Dynasties.
- Some kanji have multiple on' yomi due to these varied influences, used in different contexts.
- Example: 明 has readings like めい ("mei") and みょう ("myō") based on context.

明

According to Kanjidic this character has the following readings:

- ON: メイミョウミン
- KUN: あ.かりあか.るいあか.るむあか.らむあき.らかあ.ける -あ.け
あ.くあ.くるあ.かす
- Names: あきらあけあすきらけさやさやかとしはるみめ

<https://www.edrdg.org/cgi-bin/wwwjdic/wwwjdic?1B>

Chinese and Korean have one reading each *ming2* and *myeong*

Character Complexity and Memory Load

- Japanese kanji characters are often complex and require memorization of many details:
 - A standard-educated adult in Japan knows about 2,000 kanji.
 - Each kanji has multiple readings (pronunciations), which change based on context and word combinations.
- Learning kanji is time-intensive, requiring years of study.
- Even native speakers may struggle with less common kanji or readings, leading to a heavy memory load.

Historical Reforms and Modern Challenges

- Orthographic reforms have been implemented to simplify kanji, but challenges remain:
 - The **Tōyō Kanji** reform (1946) and **Jōyō Kanji** list (1981) aimed to standardize kanji usage.
 - However, non-standard or rare kanji still appear in literature and names, requiring additional study.
- Modern technology poses new issues:
 - Typing kanji requires selecting the correct character from multiple homophones.
 - Many younger people may rely on digital aids, impacting kanji writing skills.
- Japanese orthography is still a challenging system to master.

Writing in Vietnam

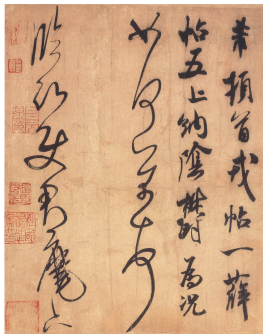
- Vietnamese today uses the Latin script-based Vietnamese alphabet to represent native Vietnamese words (thuần Việt)
- From 111 BC up to the 20th century, Vietnamese literature was written in Văn ngôn (Classical Chinese) using chữ Hán (Chinese characters)
- From the 13th century to 20th century it could also be written in Nôm (Chinese and original Vietnamese characters adapted for vernacular Vietnamese)
- From the 17th century Portuguese missionaries developed a script based on Portuguese, which was mainly used within the Catholic community. During the French administration, the alphabet was further modified and then later made a part of compulsory education in 1910

Thuần Việt

- 29 letters: 7 letters use diacritics: <ă>, <â>, <ê>, <ô>, <ơ>, <ư>, and <đ>
- An additional 5 diacritics used to designate tone (as in <à>, <á>, <ả>, <ã>, and <ạ>), level tone is unmarked.
- Diacritics stack, e.g. *nhất* “first”
- Sound changes in the spoken language have led to different letters, digraphs and trigraphs now representing the same sounds.
 - Northern dialect has 6 tones
 - Southern dialect has merged <à> *hỏi* and <ã> *ngã* to just <ả> *hỏi*.

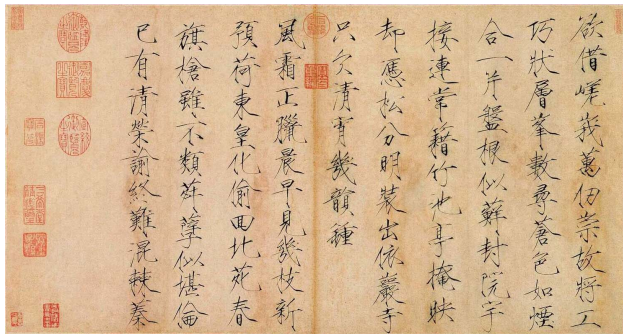
Calligraphy in East Asia

- Highly valued art form, more esteemed than painting.
- Reflects cultural aesthetics and individuality.
- Styles include Seal Script, Clerical Script, Standard Script, Running Script, and Grass Script.
- Chinese calligraphy emphasizes brush techniques and stroke order.



Mi Fu (1051–1107)

Francis Bond



Emperor Huizong of Song (1082–1135)

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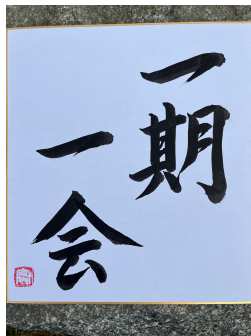
- Emphasis on the beauty of handwriting.
- Grass style and Zen calligraphy notable for their fluidity and abstraction.



別無工夫

betsu-naku kuufuu

nothing else but focus



一期一会

ichi-go ichi-e

one lifetime one chance

Indonesian Calligraphy

- Jawi calligraphy is based on Arabic calligraphy
- Often very ornate



Arabic calligraphy in Diwani style



يَا أَيُّهَا النَّبِيُّ إِنَّا أَرْسَلْنَاكَ شَاهِدًا وَمُبَشِّرًا وَنَذِيرًا وَدَاعِيًا إِلَى اللَّهِ

”O Prophet, indeed We have sent you as a witness and a bringer of good tidings and a warner, and as one who invites to Allah.”

Mehmet Izzet al-Karkuki (1841-1904)

Diwani is a calligraphic variety of Arabic script, developed during the reign of the early Ottoman Turks (16th century - early 17th century).

Summary

- There are many, many different scripts, with adaptations and borrowings
- It is very hard to change a nation's script although it does happen
 - Most countries have abandoned or limited the use of Chinese Characters
 - But having to learn a new script and reprint everything is a massive undertaking, ...

Acknowledgements

- Images taken from Omniglot (<https://omniglot.com/>) unless otherwise stated